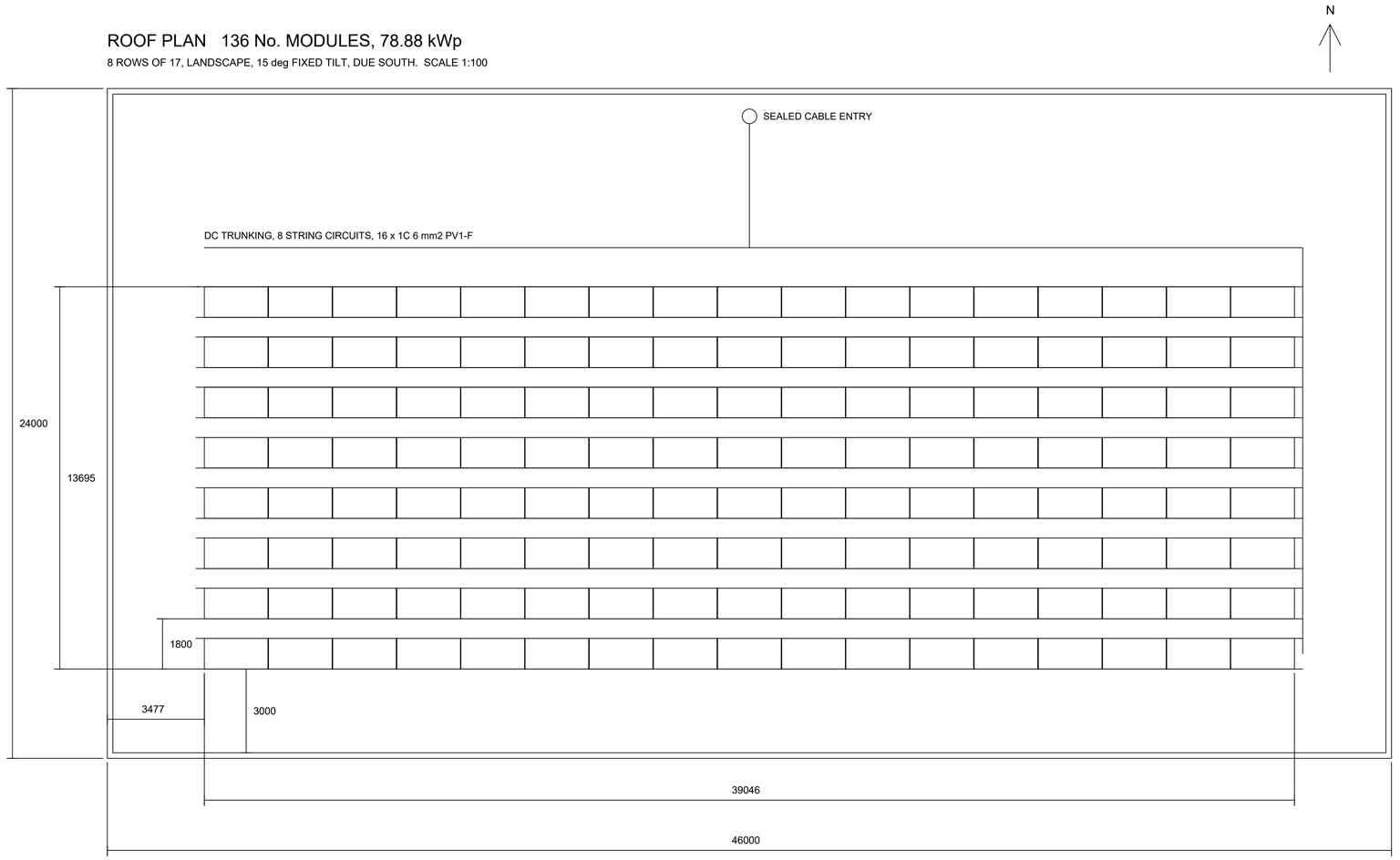
ROOF PLAN 136 No. MODULES, 78.88 kWp

8 ROWS OF 17, LANDSCAPE, 15 deg FIXED TILT, DUE SOUTH. SCALE 1:100



MECHANICAL NOTES

1. ARRAY 136 No. JINKO JKM580N-72HL4-V, 2278 x 1134 x 30 mm, LANDSCAPE,
   15 deg FIXED TILT, DUE SOUTH, IN 8 ROWS OF 17.
   ONE ROW IS ONE STRING AND FEEDS ONE MPP TRACKER OF ITS OWN.
2. ROW PITCH IS SET BY THE SHADING CASE, NOT BY A RULE OF THUMB.
   ON 21 DECEMBER AT 09:00 THE SUN IS AT 18.6 deg ELEVATION AND
   136.8 deg AZIMUTH. THE 294 mm BACK EDGE THEN CASTS A 635 mm SHADOW.
   ON TOP OF THE 1095 mm PROJECTED MODULE DEPTH THAT GIVES 1731 mm
   MINIMUM, ROUNDED UP TO THE 1800 mm PITCH DRAWN.
3. GROUND COVER RATIO 0.63. ARRAY FOOTPRINT 39046 x 13695 mm,
   WHICH IS 534.7 m2 ON A 1104 m2 ROOF, SO THE ARRAY ENVELOPE TAKES 48.4 % OF IT.
4. MOUNTING IS BALLASTED. NO ROOF PENETRATION FOR THE FRAME.
   FRONT RAIL 300 mm AND REAR RAIL 594 mm ABOVE THE MEMBRANE.
5. 3000 mm CLEAR TO THE SOUTH PARAPET AND A CLEAR WALKWAY ON THE
   NORTH SIDE FOR ACCESS AND FOR THE DC TRUNKING RUN.
6. THE ONLY ROOF PENETRATION IS THE SEALED CABLE ENTRY SHOWN.
7. BALLAST MASS, UPLIFT AND THE ADDED DEAD LOAD ON THE ROOF ARE TO BE
   CONFIRMED BY A SITE WIND LOAD AND STRUCTURAL ASSESSMENT BEFORE
   INSTALLATION. THE FIGURES ON THIS SHEET ARE DESIGN FIGURES.

EQUIPMENT SCHEDULE

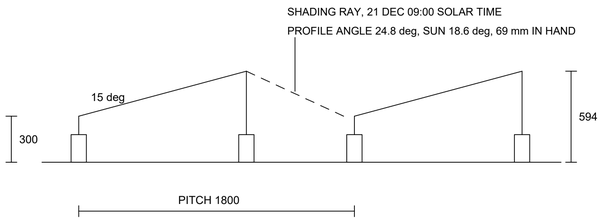
No.	EQUIPMENT	QTY	NOTE
1	JINKO JKM580N-72HL4-V 580 W	136	2278 x 1134 x 30
2	FOXESS P3-80-Plus INVERTER	1	80 kW, 8 MPPT
3	FOXESS CQ7 BATTERY MODULE	9	63.18 kWh TOTAL
4	BALLASTED TILT FRAME 15 deg	8 ROWS	FRONT 300, REAR 594
5	DC TRUNKING AND CABLE ENTRY	1 RUN	SEALED, SURFACE RUN

DESIGN SUMMARY

ARRAY	136 x 580 W = 78.88 kWp
STRINGS	8 STRINGS OF 17, ONE PER MPP TRACKER
TILT AND PITCH	15 deg, 1800 mm, GCR 0.63
INVERTER	FOXESS P3-80-Plus, 80 kW, DC/AC 0.99
BATTERY	63.18 kWh, 41.5 kW CONTINUOUS, 518.4 V NOMINAL
ARRAY ENVELOPE	534.7 m2 OF 1104 m2, 48.4 %

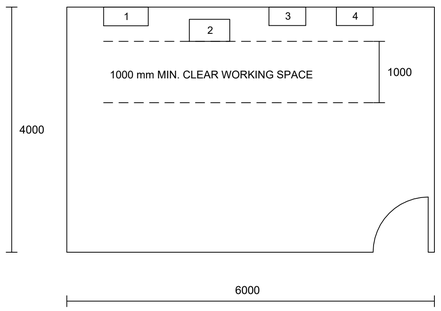
DETAIL A MOUNTING SECTION

SCALE 1:20, SOUTH ON THE LEFT



DETAIL B PLANT ROOM

SCALE 1:50



1 FOXESS P3-80-Plus HYBRID INVERTER, 730 x 305, WALL MOUNTED
2 FOXESS CQ7 BATTERY, 9 MODULES, 660 x 360 FOOTPRINT
3 AC PANEL WITH 160 A 4 POLE MCCB
4 EPS BOARD, ESSENTIAL LOADS
DOOR 900 mm, OPENING IN, AT THE SOUTH EAST CORNER

80 kW ROOFTOP PV WITH 63 kWh BATTERY STORAGE

MECHANICAL LAYOUT AND ARRAY SETTING OUT

DRAWN USAMA JAVED SITE I-10 INDUSTRIAL AREA, ISLAMABAD	SCALE 1:100 @ A1, DETAILS AS NOTED DWG PV-01
REV P1 DATE 08.10.2026 CHK -- APP --	ISSUED FOR COMMENT